

Plastic Waste To Renewable Energy—Implementing A Circular Economy

NGOs, investors, waste management practitioners and environmental conservation groups should recognise waste plastics as a resource for a clean renewable energy, and join hands with the plastic-to-fuel innovators to develop more pioneering technologies that produce ultra-clean emissions and replace fossil fuels



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“Nowadays people know the price of everything and the value of nothing.”

—**Oscar Wilde**

The archaeological timeline of human evolution shows how we have organised ourselves through the Stone Age, Bronze Age, Iron Age, and Polymer Age. These ages have proved that we can blend and amend natural resources and turn them into desirable materials. In this Polymer Age, we are showcasing that apart from blending and amending, we can manipulate the structure of chemicals and create artificial products. The versatility of plastics finds tremendous application in all household and consumer products, thus playing a major role in domestic and industrial areas.

The growth in use of plastics in products ends up with wastes that are

non-biodegradable, thus becoming an environmental hazard. Plastics with polystyrene and polypropylene become a major problem because these cannot be recycled. Even when means to recycle them are found, their saleable value being very little, recyclers throw these plastics into landfills. As plastics never decompose, the landfills become a major environmental concern.

Plastic debris is also polluting the oceans and has become a major detrimental factor to vulnerable aquatic lives. Ellen MacArthur Foundation, in partnership with the World Economic Forum, has estimated that by 2050 oceans will have more plastics than fish. We, as global citizens, need not become extremists and call for an absolute close to plastic production. Let us understand that plastic is not the problem but its way of disposal is the main concern. Actually, we desperately need a paradigm shift in perceiving things. **Let us be aware that plastics are a solid form of oil. So let us keep the plastics in the supply chain and work together to break the misperception of plastic wastes, which can be a valuable sustainable fuel.**

Why is WTE unsuccessful in India?

Waste-to-energy (WTE) plants installed in the USA and European cities have now made them the most sustainable and eco-friendly cities. The solid wastes used in these plants are materials that cannot be recycled, but the plants recover the energy from these wastes either in the form of heat or electricity. When these cities have proved a proportional relationship between WTE plants and sustain-



Fig. 1: IIP's products from waste plastics (Credit: CSIR-IIP)